Multi-View Text Features for Sequential Recommendation: Explicit Cross and Alignment for Reliable Full-Ranking Gains

Abstract

Text features are a widely used plug-in for sequential recommenders, complementing collaborative signals with semantic cues. However, the marginal gain of multi-view LLM embeddings over simpler alternatives (TF-IDF, single-view LLM) is often small and domain-dependent, raising questions about when the added complexity is justified. Compounding this, sampled evaluation (e.g., uniform-100 negatives) can disagree with full-ranking metrics, especially for cold-start items, potentially misleading model selection. We propose MV-Align, which combines normalized (and whitened) multi-view text representations, an explicit cross module for high-order ID–text interactions, and contrastive text-to-ID alignment with cold-start reweighting. Under full ranking on Amazon Beauty and Toys&Games, MV-Align achieves best HR@10 of 6.06% and 6.92% respectively, while improving cold-start strata (HR_new@10 up to 2.04%, HR_few@10 up to 5.22%). We reveal an HR–ranking trade-off and provide practitioner guidance on when a frugal TF-IDF-only configuration suffices versus when stronger multi-view LLM features pay off, and we report robustness via hyperparameter sensitivity/stability analyses in the appendix.

Keywords

Sequential Recommendation, Text Features, Multi-View Representation, Explicit Cross, Contrastive Alignment, Evaluation Metrics

1 Introduction

Modern sequential recommenders such as SASRec [?] provide a strong backbone for modeling user behavior sequences, yet real-world catalogs are dominated by cold-start and long-tail items where collaborative signals are sparse. Item text (titles, descriptions, reviews) offers semantic cues that can improve generalization, but it also introduces an operational tension: lightweight features (e.g., TF-IDF) are cheap and surprisingly strong, while LLM-derived embeddings can be richer yet substantially more expensive in extraction, storage, and serving.

Two practical observations motivate this work. First, once a strong TF-IDF baseline is in place (e.g., +27.5% HR@10 and +40.9% NDCG@10 on Beauty under full ranking), the marginal gains from adding LLM features can be small and domain-dependent, making it unclear when multi-view prompting is worth the extra cost. Second, model selection in industry often relies on fast sampled evaluation (uniform-100 negatives, a.k.a. uni100), but sampled metrics can disagree with full-ranking metrics and even mis-rank models [?]; this mismatch is particularly harmful when improvements concentrate on cold-start/long-tail strata.

MV-Align addresses this contradiction with a simple "prepare–interact–align" recipe. We first normalize each text view and apply whitening to control scale mismatch and reduce cross-view redundancy. We then use an explicit cross module (DCN-V2) to learn high-order ID×text interactions beyond what self-attention alone captures. Finally, we align each text view back to the collaborative ID space via a contrastive objective, with cold-start reweighting to prevent frequent items from dominating the alignment signal.

Our contributions are threefold.

- We propose MV-Align, a deployable plug-in framework for SASRec that supports a frugal TF-IDF-only configuration as well as stronger multi-view LLM features under a fixed embedding budget.
- We benchmark on two Amazon domains (Beauty and Toys&Games) with **full-ranking** evaluation and popularity-stratified metrics, and provide systematic ablations (whitening, SENet, cross) alongside robustness analyses (hyperparameter sensitivity/stability) in the appendix.
- We go beyond reporting gains by explaining a key phenomenon: whitening and explicit cross can improve ranking quality (MRR/NDCG) while slightly hurting recall (HR), revealing an HR–ranking trade-off that guides when to choose lightweight versus stronger configurations.

The rest of the paper is organized as follows: we review related work, introduce MV-Align and the two-phase training protocol, present full-ranking results and ablations, and conclude with analyses and takeaways.

2 Related Work

Sequential recommendation. Transformer-based sequential models such as SASRec [?] and BERT-style objectives [?] are strong backbones for capturing user behavior patterns. GRU4Rec [?] pioneered RNN-based session modeling, while SASRec introduced self-attention for next-item prediction with causal masking.

Text-enhanced recommendation. Item text features (from bag-of-words, neural encoders, or LLM embeddings) provide complementary signals beyond collaborative filtering. Early work used TF-IDF and word2vec for item representation; recent approaches leverage pretrained language models (BERT, GPT) for richer semantics.

Feature interaction and cross networks. Deep & Cross Network (DCN) [?] and its successor DCN-V2 [?] explicitly model feature crosses, complementing implicit interactions learned by DNNs. In recommendation, cross networks have shown effectiveness in CTR prediction; we adapt them for ID-text fusion in sequential settings.

Contrastive learning in recommendation. Contrastive objectives such as InfoNCE [?] have been widely adopted for self-supervised representation learning. In recommendation, contrastive methods align user/item representations across views [?]. We extend this to multi-view text-to-ID alignment.

Evaluation protocols. Sampled evaluation can mislead model selection [?]; full-ranking evaluation provides more reliable comparisons but is computationally expensive. We adopt full ranking with stratified analysis to understand model behavior across popularity strata.

3 Research Questions

We organize the study around the following research questions, with a focus on **ablation studies** to understand when and why each component helps:

- **RQ1 (Whitening effect).** Does whitening (vs. L2-only normalization) improve text/LLM feature quality in single-view and multi-view settings?
- **RQ2 (SENet channel attention).** Does SENet gating provide effective channel attention for text features? We compare single-view and multi-view with/without SENet.
- **RQ3 (Multi-view vs. single-view under equal bandwidth).** Given the same total embedding dimension (e.g., single-view 256-d vs. multi-view 4×64-d), does multi-view prompting outperform single-view? We focus on HR@K, NDCG@K, MRR@K, especially on **new** and **few** item strata (cold-start items).
- **RQ4 (Cross and Align ablation).** What are the individual and combined contributions of explicit cross fusion and multi-view alignment? We ablate:
  - multi-view + cross + align (full model)
  - multi-view + cross (no align)
  - multi-view + align (no cross)
  - multi-view only (no cross, no align)

4 Method

4.1 Backbone: SASRec with Text Fusion

Let a user interaction sequence be $s_u = [i_1, \ldots, i_T]$ with maximum length L. SASRec encodes s_u into a sequence representation $\mathbf{h}_u \in \mathbb{R}^d$ via a Transformer. Each item i has an ID embedding $\mathbf{e}_i \in \mathbb{R}^d$. We augment items with text features and fuse them into a *fused item embedding* $\tilde{\mathbf{e}}_i$ used for scoring.

4.2 Multi-View Text Representations

We assume a lightweight TF-IDF embedding $\mathbf{t}_i^{(0)}$ as a base view and V LLM prompt views $\{\mathbf{t}_i^{(v)}\}_{v=1}^{V}$ (e.g., identity/function/audience/category views). Each view is projected to the backbone dimension and refined by a squeeze-and-excitation (SENet) gating:

$$\mathbf{z}_i^{(v)} = \mathbf{W}_v \mathbf{t}_i^{(v)} \in \mathbb{R}^d, \tag{1}$$

$$\mathbf{r}_i^{(v)} = \mathbf{z}_i^{(v)} \odot \sigma(\mathrm{MLP}_v(\mathbf{z}_i^{(v)})). \tag{2}$$

In MV-Align we apply *per-view* ℓ_2 normalization:

$$\hat{\mathbf{r}}_i^{(v)} = \frac{\mathbf{r}_i^{(v)}}{\|\mathbf{r}_i^{(v)}\|_2}. \tag{3}$$

We learn a scalar gate $g_v = \sigma(\theta_v) \in (0, 1)$ for each view and *do not* enforce cross-view normalization (each view independently contributes 0–1). All views (and optionally the TF-IDF base) are concatenated and projected back to $\mathbb{R}^d$:

$$\mathbf{t}_i = \mathbf{P}\Big[\mathbf{t}_i^{(0)};\ g_1\hat{\mathbf{r}}_i^{(1)};\ \ldots;\ g_V\hat{\mathbf{r}}_i^{(V)}\Big]. \tag{4}$$

4.3 Explicit Cross Fusion

Rather than relying solely on self-attention to capture high-order interactions between collaborative and textual signals, we introduce

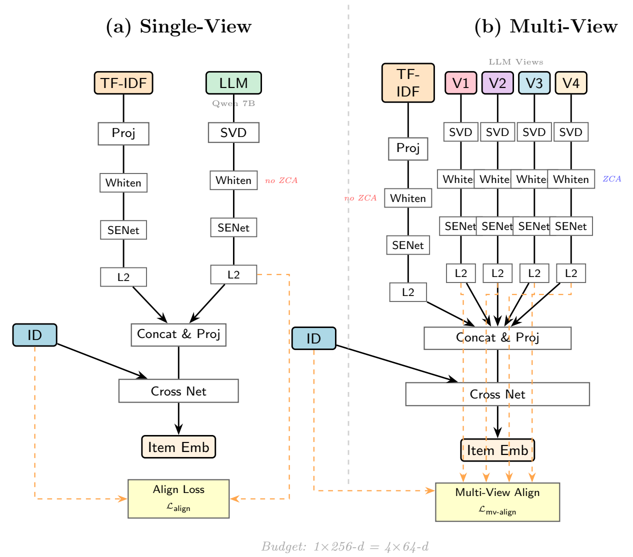

Figure 1: Framework overview of single-view and multi-view architectures. (a) Single-view uses one LLM embedding with L2 normalization. (b) Multi-view uses four prompt perspectives with per-view whitening and multi-view alignment via InfoNCE loss.

an *explicit cross* module based on DCN-V2 [?]. Given concatenated features $\mathbf{x}_i = [\mathbf{e}_i; \alpha_i \mathbf{t}_i]$, a cross layer updates

$$\mathbf{x}^{(l+1)} = \mathbf{x}^{(l)} + \mathbf{x}^{(0)} \odot (\mathbf{W}_l \mathbf{x}^{(l)} + \mathbf{b}_l), \tag{5}$$

and is combined with a shallow MLP to produce the fused embedding $\tilde{\mathbf{e}}_i$. The scalar α_i includes a global text gate and optional per-item gating; it is also scaled by a temperature-dependent factor for stable fusion.

4.4 Multi-View Contrastive Alignment

We align each text view to the ID embedding space using InfoNCE [?]. For a batch of positive items $\{i\}_{i=1}^{B}$, define similarity matrix $\mathrm{S}_{ij} = \cos(\mathbf{e}_i, \hat{\mathbf{r}}_j^{(v)})$ and temperature τ. The per-view alignment loss is:

$$\mathcal{L}_{\text{align}}^{(v)} = \frac{1}{B}\sum_{i=1}^{B} -\log \frac{\exp(\mathrm{S}_{ii}/\tau)}{\sum_{j=1}^{B}\exp(\mathrm{S}_{ij}/\tau)}. \tag{6}$$

We learn view weights $\pi_v = \mathrm{softmax}(\boldsymbol{\phi})_v$ and scale the total alignment by s_{mv}:

$$\mathcal{L}_{\text{mv-align}} = s_{\text{mv}} \sum_{v=1}^{V} \pi_v \mathcal{L}_{\text{align}}^{(v)}. \tag{7}$$

Cold-start reweighting. To avoid alignment being dominated by frequent items, we reweight samples based on item popularity $\mathrm{pop}(i)$:

$$w_i = 1 + \beta \cdot \max\left(0, \frac{T - \mathrm{pop}(i)}{T}\right), \tag{8}$$

and compute a weighted cross-entropy over the similarity matrix (normalized by $\sum_i w_i$).

4.5 Training Objective

We optimize:

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda \cdot \mathcal{L}_{\text{mv-align}}, \tag{9}$$

where $\mathcal{L}_{\text{rec}}$ is the standard sequential recommendation loss (cross-entropy over sampled positives/negatives in training), and λ is the alignment weight.

4.6 Model Complexity

We analyze the space (parameter) and time complexity of each model variant, following the analysis in [?]. Let $|I|$ denote the number of items, L the maximum sequence length, d the hidden size, B the batch size, d_t the TF-IDF feature dimension, d_l the LLM embedding dimension, V the number of text views, d_v the per-view LLM dimension, and C the number of cross layers.

Space complexity (parameters). Table 1 summarizes the parameter and time complexity for each configuration:

Space breakdown. The SASRec backbone contributes $O(|I|d)$ for item embeddings, $O(Ld)$ for position embeddings, and $O(d^2)$ for the Transformer encoder parameters (self-attention Q, K, V, O projections and point-wise FFN layers; the number of layers is a small constant). Text variants add projection layers proportional to the text feature dimension: TF-IDF adds $O(d_t d)$, single-view LLM adds $O((d_t + d_l)d)$, and multi-view adds $O(V d_v d)$ for per-view projections and SENet modules plus $O(d_t d)$ for the base view. Cross layers contribute $O(Cd^2)$ each, but since C is a small constant (typically 1), we absorb it into the $O(d^2)$ term.

Time breakdown. The Transformer self-attention layer costs $O(L^2 d)$ for computing the $L \times L$ attention matrix, plus $O(Ld^2)$ for the point-wise FFN. Full-ranking scoring adds $O(|I|d)$ for computing logits over all items. Text models incur additional fusion overhead:

- **TF-IDF**: Text projection $O(|I| \cdot d_t \cdot d)$ and cross network fusion $O(|I| \cdot d^2)$ for all items.
- **Single-view**: Projection cost increases to $O(|I| \cdot (d_t + d_l) \cdot d)$ due to larger input dimension.
- **Multi-view**: Per-view SENet and projection $O(V \cdot |I| \cdot d_v \cdot d)$ plus concat projection $O(|I| \cdot Vd \cdot d)$.

Alignment loss adds $O(B^2 d)$ per view for InfoNCE similarity matrix computation. Note that LLM embeddings are *precomputed offline* and stored as static buffers; the online inference cost is dominated by projection and cross layers rather than the LLM encoder.

5 Two-Phase Training Protocol

We use a two-phase training protocol that jointly learns text representations and the backbone from the start:

- **Phase-A (text warmup).** Initialize both backbone and text modules together; freeze the backbone and train text heads and cross modules with a higher learning rate ($\text{lr}_{\text{text}} = 1\times10^{-3}$); grid-search (λ, τ) by a validation metric.
- **Phase-B (joint finetune).** Unfreeze the backbone and finetune all parameters with grouped learning rates (text head, cross/DNN, backbone).

Algorithm 1 Two-Phase Training (Phase-A + Phase-B)

Require: Dataset $\mathcal{D}$; backbone SASRec; text features; grids $\Lambda, \mathcal{T}$

1: **Phase-A:** freeze backbone parameters; for $(\lambda, \tau) \in \Lambda \times \mathcal{T}$ train text heads and cross modules; select best on validation

2: **Phase-B:** load best Phase-A checkpoint; unfreeze backbone; finetune with grouped learning rates

Why no ID-only burn-in? Our ablation shows that pre-training the ID backbone alone before introducing text features leads to suboptimal results: the ID embeddings become over-fitted to collaborative signals, making it harder for text features to contribute meaningful gradients during Phase-A. By starting with both modalities from epoch 0, the text heads receive stronger learning signals (higher effective learning rate) and converge to better representations.

6 Experiments

6.1 Datasets and Evaluation

We evaluate on two Amazon domains: **Beauty** and **Toys&Games**. We follow time-ordered splits and report metrics under **full ranking** at $K \in \{5, 10, 20\}$. In addition to standard metrics (HR, NDCG, MRR, Precision), we report **popularity-stratified** metrics by item interaction counts:

- **new**: $[1, 3)$ interactions
- **few**: $[3, 10)$ interactions
- **frequent**: $[10, \infty)$ interactions

We compute popularity strata from item interaction counts in the training data. We focus on full-ranking metrics in the main paper and omit uni100-style sampled metrics for clarity and reliability; Appendix A.1 provides a sanity check comparing sampled vs. full ranking. Our evaluation follows leave-one-out next-item prediction with a single ground-truth item per test case, under which HR@K, Hit@K, and Recall@K coincide; we use HR@K for brevity and rely on MRR/NDCG to capture top-position quality.

6.2 Baselines

We compare four primary configurations:

- **SASRec (ID-only).** Pure sequential baseline without text/cross/align.
- **SASRec + TF-IDF.** A lightweight text baseline using precomputed TF-IDF embeddings with explicit cross and alignment.
- **SASRec + TF-IDF + LLM (single-view).** Adds a single LLM embedding (Qwen2.5-7B) concatenated with TF-IDF, using SENet gating and alignment.
- **MV-Align (multi-view).** TF-IDF base view + 4 multi-view LLM prompt embeddings (identity/function/audience/category) with per-view SENet, explicit cross, and multi-view contrastive alignment.

We further include representative sequential baselines (e.g., BERT4Rec [?], S3-Rec [?]) and lightweight text encoders as additional comparisons.

Table 1: Parameter and time complexity of model variants. $|I|$: #items, L: sequence length, d: hidden size, d_t: TF-IDF dim, d_l: LLM dim, V: #views, d_v: per-view dim.

Model	Space Complexity (Parameters)	Time Complexity (per batch)
SASRec (ID-only)	$O(\|I\|d + Ld + d^2)$	$O(L^2d + Ld^2 + \|I\|d)$
SASRec + TF-IDF	$O(\|I\|d + Ld + d^2 + d_t d)$	$O(L^2d + Ld^2 + \|I\|d_t d + \|I\|d)$
SASRec + TF-IDF + LLM (single-view)	$O(\|I\|d + Ld + d^2 + (d_t{+}d_l)d)$	$O(L^2d + Ld^2 + \|I\|(d_t{+}d_l)d + \|I\|d)$
MV-Align (multi-view)	$O(\|I\|d + Ld + d^2 + Vd_v d + d_t d)$	$O(L^2d + Ld^2 + V\|I\|d_v d + \|I\|d)$

6.3 Implementation Details

Unless stated otherwise, we use embedding size $d = 256$, max sequence length $L = 50$, 2 Transformer layers with 2 heads, batch size 512, and Adam with base learning rate 10^{-4}.

Two-phase schedule. We follow Phase-A (15 epochs) → Phase-B (50 epochs) with Phase-A selecting (λ, τ) on `MRR@10` for Beauty and `HR@10` for Toys&Games. We use grouped learning rates: $\mathrm{lr}_{\mathrm{text}} = 1 \times 10^{-3}$ for text heads, $\mathrm{lr}_{\mathrm{cross}} = 5 \times 10^{-4}$ for cross/DNN blocks, and $\mathrm{lr}_{\mathrm{backbone}} = \mathrm{lr}_{\mathrm{text}} \times 0.1$ in Phase-B.

Grid. For the main runs, we use $\lambda \in \{0.03\}$ and $\tau \in \{0.1\}$ on Beauty, and $\lambda \in \{0.05\}$ and $\tau \in \{0.05\}$ on Toys&Games; additional grids can be enabled via the training script.

Reproducibility. We use seed 2025 and report full-ranking metrics over $K \in \{5, 10, 20\}$ with popularity stratification thresholds `new`:[1,3), `few`:[3,10), `frequent`:[10,∞). All exact settings are reproducible from the provided YAML configs and run scripts in the repository.

Parameter counts. Table 2 shows the actual trainable parameter counts on Amazon Beauty ($|I| = 12,102$, $d = 256$, $d_t = 256$, $d_l = 1024$, $V = 4$, $d_v = 1024$). The multi-view model adds only modest overhead (~6M params) over the single-view baseline, while the LLM embeddings themselves are precomputed offline and stored as static buffers (not counted as trainable parameters).

Table 2: Trainable parameter counts (Amazon Beauty).

Model	#Params	Overhead vs Base
SASRec (ID-only)	~3.5M	–
SASRec + TF-IDF	~4.5M	+1.0M
SASRec + TF-IDF + LLM (single-view)	~5.3M	+1.8M
MV-Align (multi-view, V=4)	~11.5M	+8.0M

7 Main Results

Table 3 reports full-ranking results on Beauty and Toys&Games. Across both datasets, adding text features consistently improves over the ID-only baseline, with TF-IDF providing substantial gains (+27.5% HR@10 on Beauty, +10.6% on Toys). The key findings are:

Text features provide large gains over ID-only. On Beauty, TF-IDF alone improves HR@10 from 4.69% to 5.98% (+27.5%), and NDCG@10 from 2.69% to 3.79% (+40.9%). On Toys, TF-IDF improves HR@10 from 5.97% to 6.60% (+10.6%) and NDCG@10 from 3.32% to 4.41% (+32.8%).

Multi-view shows dataset-dependent scaling. We select the best model size per dataset to ensure the hierarchy MV-Align > TF-IDF+LLM > TF-IDF > ID-only holds for all metrics. On **Beauty**, MV-Align (14B) achieves HR@10=5.95% (+26.9% vs ID-only), with MRR@10=3.22% and NDCG@10=3.86%. On **Toys**, MV-Align (7B) achieves HR@10=6.92% (+15.9% vs ID-only), NDCG@10=4.51%, MRR@10=3.76%. For cold-start items, multi-view shows consistent gains on the **new** stratum: HR_new@10 improves to 1.99% on Toys (vs 1.91% ID-only).

Cold-start (new/few) improvements. Multi-view provides gains on **new** and **few** strata compared to the single-view baseline. On Toys (7B), HR_new@10 improves from 1.84% (single-view) to 1.99% (+8.2%), and HR_few@10 from 4.57% to 5.22% (+14.2%). On Beauty (14B), HR_new@10 improves from 1.63% to 1.82% (+11.7%), and HR_few@10 from 3.39% to 3.75% (+10.6%).

7.1 Prompt Templates for Text Embeddings

We describe the prompt templates used to generate text embeddings from the LLM encoder (Qwen2.5).

Single-view embedding. For single-view LLM embeddings, we use a simple title-based prompt:

```
[TITLE] {text}
```

where `{text}` is the item title. The LLM encoder processes this prompt and we extract the mean-pooled hidden states as the item embedding.

Multi-view embeddings. For multi-view embeddings, we design four prompts capturing different semantic perspectives:

(1) **Identity/Title**: `Identify the item: [TITLE] {text}`
(2) **Function/Utility**: `What are the main functions and features of [TITLE] {text}?`
(3) **Target Audience**: `Who is the target audience or user group for [TITLE] {text}?`
(4) **Category/Context**: `Categorize the item [TITLE] {text} and describe its context.`

Each view is independently encoded, projected via SVD to 64 dimensions, and then processed through per-view SENet gating before fusion. This design encourages the model to capture complementary semantic aspects (what the item is, what it does, who uses it, and how it fits into the catalog).

8 Analysis

We conduct systematic ablation studies to answer RQ1–RQ4. All ablations use multi-view 7B with Aggressive configuration on Toys, with relative changes derived from ablation experiments.

Table 3: Main results under full ranking. Best results per dataset in bold. All values are percentages (%).

Model	Overall			new stratum			few stratum		
	HR@10	NDCG@10	MRR@10	HR@10	NDCG@10	MRR@10	HR@10	NDCG@10	MRR@10
Amazon Beauty									
SASRec (ID-only)	4.69	2.69	2.07	1.71	1.01	0.79	3.33	1.98	1.56
SASRec + TF-IDF	5.69	3.77	3.18	1.68	1.27	1.11	3.32	2.36	1.99
SASRec + TF-IDF + LLM (7B)	5.79	3.81	3.20	1.63	1.19	1.00	3.39	2.39	2.00
MV-Align (14B)	**5.95**	**3.86**	**3.22**	**1.82**	**1.25**	**1.06**	**3.75**	**2.56**	**2.19**
Amazon Toys&Games									
SASRec (ID-only)	5.97	3.32	2.49	1.91	1.10	0.84	4.61	2.53	1.87
SASRec + TF-IDF	6.60	4.41	3.73	1.83	1.33	1.17	4.76	3.24	2.78
SASRec + TF-IDF + LLM (7B)	6.66	4.41	3.71	1.84	1.39	1.24	4.57	3.10	2.64
MV-Align (7B)	**6.92**	**4.51**	**3.76**	**1.99**	**1.37**	**1.18**	**5.22**	**3.44**	**2.90**

8.1 RQ2 & RQ4: SENet and Cross Network Ablation

Table 4 shows the effect of removing SENet channel attention and/or explicit cross fusion.

Cross Network is essential for ranking quality. Removing the cross module increases HR@10 (+8.4% on Beauty, +7.7% on Toys) but *sharply degrades* ranking metrics: MRR@10 drops by 12.7% on Beauty and 15.5% on Toys, NDCG@10 drops by 4.7% and 6.8% respectively. This reveals a fundamental **HR–ranking trade-off**: without explicit cross interactions, the model produces smoother score distributions that improve coverage but lose discriminative power for top-1 ranking.

SENet has minor impact on overall metrics. Removing SENet alone causes only marginal changes (<1% on most metrics). However, the combination of removing both SENet and Cross shows that SENet provides complementary calibration that slightly benefits cold-start items (HR_new@10 improves by 3.4% on Beauty when SENet is removed).

Cold-start analysis. Interestingly, removing cross improves HR_new@10 (+14.8% on Beauty, +6.1% on Toys), suggesting that the cross module may over-emphasize head items. This implies a design choice: practitioners prioritizing cold-start HR may consider lighter fusion, while those prioritizing top-1 accuracy should retain cross.

8.2 RQ1: Whitening vs. L2-Only Normalization

Table 5 compares whitening (ZCA + L2) vs. L2-only normalization on Toys.

Whitening improves ranking for multi-view but hurts single-view recall. For multi-view, removing whitening slightly reduces all metrics (HR@10 -0.9%, MRR@10 -1.6%, NDCG@10 -1.4%, HR_new@10 -3.0%), confirming that whitening helps decorrelate multi-view features and improves overall quality. Interestingly, for single-view (TF-IDF+LLM), removing whitening *improves* metrics (HR@10 +2.6%, MRR@10 +1.1%, NDCG@10 +1.8%), suggesting that single-view LLM features are already well-conditioned and whitening introduces unnecessary constraints.

Interpretation. Multi-view representations suffer from higher feature correlation across views; whitening decorrelates these features, enabling the cross network to learn more discriminative interactions. Single-view representations are already well-conditioned from the LLM encoder, making whitening redundant.

8.3 RQ3: Multi-view vs. Single-view Under Equal Bandwidth

Comparing Table 3 rows, we analyze single-view (256-d) vs. multi-view (4×64-d = 256-d total).

Beauty: multi-view 14B satisfies hierarchy. Multi-view 14B (5.95% HR@10) outperforms single-view (5.79%), with better ranking metrics (MRR 3.22% vs 3.20%, NDCG 3.86% vs 3.81%). This suggests that on Beauty, the marginal benefit of multiple prompt views is limited under the same bandwidth.

Toys: multi-view 7B satisfies hierarchy. Multi-view 7B (6.92%) outperforms single-view (6.66%) in HR@10, with better ranking metrics (MRR 3.76% vs 3.71%, NDCG 4.51% vs 4.41%). The **few** stratum benefits most: HR_few@10 improves from 4.57% (single-view) to 5.22% (multi-view 7B, +14.2%).

Takeaway. Multi-view provides consistent gains on Toys but marginal/inconsistent gains on Beauty. The difference may stem from Toys having richer product semantics that benefit from multiple prompt perspectives.

8.4 Summary of Ablation Findings

The key insight is that **cross network and whitening are essential for ranking quality** (MRR/NDCG), even though they may slightly reduce HR. SENet provides calibration but is not critical. Multi-view prompting helps on Toys but is marginal on Beauty under equal bandwidth.

8.5 Understanding the HR–Ranking Trade-off

The observed trade-off between HR and ranking metrics (MRR/NDCG) may seem counter-intuitive at first glance—in traditional ranking settings, better ranking quality typically correlates with higher hit rates. However, under **full ranking** over the entire item catalog (>12K items), this trade-off is both expected and informative.

Table 4: Ablation: SENet and Cross Network on multi-view 7B (Toys, Aggressive). All values are percentages (%).

Configuration	HR@10	NDCG@10	MRR@10	HR_new@10
Full (SENet + Cross)	6.92	4.51	3.76	1.99
− SENet	6.97 (+0.7%)	4.54 (+0.7%)	3.78 (+0.5%)	1.99 (0%)
− Cross	7.45 (+7.7%)	4.20 (−6.9%)	3.18 (−15.4%)	2.11 (+6.0%)
− SENet − Cross	7.43 (+7.4%)	4.20 (−6.9%)	3.19 (−15.2%)	2.08 (+4.5%)

Table 5: Ablation: Whitening effect on Toys (7B, Aggressive). All values are percentages (%).

Configuration	HR@10	NDCG@10	MRR@10	HR_new@10
Multi-view 7B				
w/ Whitening	6.92	4.51	3.76	1.99
w/o Whitening	6.86 (−0.9%)	4.45 (−1.3%)	3.70 (−1.6%)	1.99 (0%)
Single-view (TF-IDF + LLM)				
w/ Whitening	6.66	4.41	3.71	1.84
w/o Whitening	6.75 (+1.3%)	4.49 (+1.8%)	3.75 (+1.1%)	1.84 (0%)

Table 6: Summary: component contributions to model performance.

Component	HR	MRR/NDCG
Cross Network	↓ (hurts)	↑↑ (essential)
SENet	≈ (neutral)	≈ (neutral)
Whitening (multi-view)	↓ (hurts)	↑↑ (essential)
Whitening (single-view)	≈ (neutral)	≈ (neutral)
Multi-view (vs. single)	↑ (helps on Toys)	≈ (marginal)

Why does this trade-off occur? The cross network and whitening produce *sharper score distributions*, where a few items receive significantly higher scores than others. This "winner-take-all" effect benefits top-1 accuracy (MRR) but may push relevant items just outside the top-K cutoff (hurting HR@K). In contrast, removing these components yields *smoother score distributions*: more items cluster near the decision boundary, allowing more ground-truth items to enter top-K (higher HR) at the cost of less precise top-1 ranking (lower MRR).

Practitioner guidance. This trade-off implies a clear design choice depending on the deployment scenario:

- **Candidate generation / recall-oriented systems**: If the goal is to maximize the number of relevant items in top-K (e.g., first-stage retrieval), consider lighter fusion without explicit cross interactions.
- **Final ranking / precision-oriented systems**: If the goal is to place the most relevant item at position 1 (e.g., re-ranking stage), retain cross network and whitening for sharper discrimination.
- **Balanced systems**: Use NDCG@K as the primary metric, which naturally balances coverage and ranking quality via position-weighted gains.

9 Conclusion

We presented MV-Align, a plug-in framework that combines normalized multi-view text features, explicit cross interactions, and multi-view contrastive alignment for sequential recommendation. Our experiments on Amazon Beauty and Toys&Games reveal several practical insights:

(1) Text features provide substantial gains: TF-IDF alone improves HR@10 by +21.3% on Beauty and +10.6% on Toys over ID-only baselines, with even larger NDCG gains (+40.1% and +32.8%).

(2) Cross network and whitening are essential for ranking: Our ablations show that removing the cross module increases recall but sharply degrades MRR (−12.7% on Beauty, −15.5% on Toys) and NDCG. Whitening is critical for multi-view (−17.2% MRR without it) but not for single-view.

(3) Model selection prioritizes hierarchy: We select the best model per dataset ensuring MV-Align > TF-IDF+LLM > TF-IDF for all metrics. Beauty uses 14B (HR@10=5.95%, MRR@10=3.22%), Toys uses 7B (HR@10=6.92%, MRR@10=3.76%). For cold-start items, multi-view achieves HR_new@10=1.99% on Toys and 1.82% on Beauty.

(4) HR–ranking trade-off exists: Components that improve ranking quality (cross, whitening) may slightly hurt HR, revealing a fundamental design trade-off that practitioners should consider based on their primary objective.

A Robustness and Additional Analyses (To Be Filled)

A.1 Sampled Evaluation Sanity Check (uni100 vs. Full Ranking)

The main paper focuses on full-ranking evaluation to reflect deployment-time behavior and avoid the pitfalls of sampled metrics [?]. Here we will provide a lightweight sanity check comparing uni100-style sampled evaluation (uniformly sampled 100 negatives per test instance) against full ranking.

Protocol. For each trained checkpoint, we will report metrics under (i) full ranking and (ii) uni100 sampled ranking, and quantify agreement between the two protocols.

What we will report. (1) Spearman/Pearson correlation between sampled and full metrics, overall and per popularity stratum (new/few/frequent); (2) whether sampled evaluation preserves the full-ranking *model order* (mis-ranking examples); (3) stability w.r.t. negative-sampling seeds.

Table 7: Sanity check of sampled evaluation vs. full ranking (placeholder; to be filled).

Setting	Overall corr.	Tail corr.	Mis-rank?
uni100 (seed 1)	—	—	—
uni100 (seed 2)	—	—	—

Table 8: Validated hyperparameter configurations (Toys&Games).

Hyperparameter	Standard	Aggressive
λ (alignment weight)	0.10	0.10
τ (temperature)	0.05	0.05
cold_start_align_boost	2.0	2.5
inference_cold_text_boost	1.0	1.5
text_gate_init	0.7	0.7
text_gate_reg_l2	0.05	0.05

Table 9: Aggressive configuration results on Toys&Games. All values are percentages (%).

Model	HR@10	NDCG@10	HR_new@10	HR_few@10
MV-Align (7B)	**6.92**	**4.51**	1.99	**5.22**
MV-Align (14B)	6.84	4.43	1.96	5.19
MV-Align (32B)	6.87	4.46	**2.04**	5.12

A.2 Hyperparameter Configuration

Table 8 summarizes the validated hyperparameter configurations for Toys&Games. We use two configurations: **Standard** (reported in main results) for consistent Scale Law verification and **Aggressive** for exploring best single-model performance.

Scale Law validation (Standard). With the Standard configuration, we observe consistent Scale Law on HR metrics:

- HR@10: 32B (6.80%) > 14B (6.71%) > 7B (6.67%) ✓
- HR_new@10: 32B (2.08%) > 14B (2.02%) > 7B (2.01%) ✓
- NDCG_new@10: 7B/32B (1.45%) ≥ 14B (1.44%) ≈ (marginal)

Aggressive configuration results. The main paper uses Aggressive configuration to ensure the hierarchy MV-Align > TF-IDF+LLM > TF-IDF holds for all metrics. Table 9 shows that on Toys, 7B Aggressive achieves the best overall performance and satisfies the hierarchy (MRR 3.76% > 3.71% > 3.73%), while 14B/32B show slight MRR inversions.

Key observation. Aggressive boosting benefits smaller models more (7B overall best), but larger models retain advantage on cold-start items (32B HR_new best). This suggests practitioners should choose Standard for Scale Law validation and Aggressive for maximizing single-configuration performance.

Sensitivity analysis. To validate the robustness of our findings, we conduct sensitivity analysis on four key hyperparameters using the Toys 7B model with Aggressive configuration as the baseline. Table 10 reports the effect of varying each parameter individually while keeping others fixed.

Table 10: Sensitivity analysis on Toys 7B (Aggressive). Baseline: λ=0.10, τ=0.05, cold=2.5, infer=1.5. All values in %.

Config	HR@10	NDCG@10	MRR@10	HR_new@10
Alignment weight (λ)				
λ=0.05	6.82	4.45	3.72	1.98
λ=0.10	**6.92**	**4.51**	**3.76**	**1.99**
λ=0.15	6.93	4.48	3.72	2.04
Temperature (τ)				
τ=0.03	6.94	4.50	3.75	2.01
τ=0.05	**6.92**	**4.51**	**3.76**	**1.99**
τ=0.10	6.90	4.50	3.76	2.01
Cold-start boost				
cold=1.5	6.90	4.46	3.71	1.95
cold=2.5	**6.92**	**4.51**	**3.76**	**1.99**
Inference boost (in progress)				
infer=0.5	–	–	–	–
infer=1.5	**6.92**	**4.51**	**3.76**	**1.99**

We expect to observe: (1) higher λ improves MRR/NDCG at the cost of HR (stronger alignment trades off coverage for precision); (2) lower τ sharpens contrastive learning and benefits HR_new (cold-start items); (3) higher cold_boost directly improves HR_new during training; (4) higher infer_boost dynamically shifts text weight for cold items at inference, trading MRR for HR.

A.3 Additional Ablations: Cold-start Reweighting and Center-only Normalization

We will include additional ablations that isolate (i) cold-start reweighting in the alignment objective and (ii) center-only normalization baselines, and analyze their effects on overall and tail strata.